

MTH 203 : Lecture 06

Second and Higher Order Derivatives:

- Let $f(x, y)$ be a function of two variables defined on $D \subseteq \mathbb{R}^2$
- We have defined $\frac{\partial f}{\partial x}$ and $\frac{\partial f}{\partial y}$
- We can take derivatives further.

$$\frac{\partial}{\partial x} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial x^2} \quad \left(\text{or } f_{xx} \right) \quad \frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x} \right) = \frac{\partial^2 f}{\partial y \partial x} \quad \left(\text{or } f_{xy} \right)$$

$$\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial y^2} \quad \left(\text{or } f_{yy} \right) \quad \frac{\partial}{\partial x} \left(\frac{\partial f}{\partial y} \right) = \frac{\partial^2 f}{\partial x \partial y} \quad \left(\text{or } f_{yx} \right)$$

Ex: $f(x, y) = \ln(\sqrt{x^2 + y^2}) = \frac{1}{2} \ln(x^2 + y^2)$

Show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0$

$$\frac{\partial f}{\partial x} = \frac{1}{2} \cdot \frac{2x}{x^2 + y^2} = \frac{x}{x^2 + y^2}$$

$$\frac{\partial f}{\partial y} = \frac{1}{2(x^2+y^2)} \times 2y = \frac{y}{x^2+y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = \frac{1 \cdot (x^2+y^2) - x(2x)}{(x^2+y^2)^2} = \frac{y^2 - x^2}{(x^2+y^2)^2}$$

$$\frac{\partial^2 f}{\partial y^2} = \frac{1 \cdot (x^2+y^2) - y(2y)}{(x^2+y^2)^2} = \frac{x^2 - y^2}{(x^2+y^2)^2}$$

$$\text{So, } \left[\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0 \right]$$

Ex: $f(x, y, z) = e^{3x+4y} \cos 5z$

Show that $\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0$

$$\frac{\partial f}{\partial x} = 3e^{3x+4y} \cos 5z$$

$$\frac{\partial f}{\partial y} = 4e^{3x+4y} \cos 5z$$

$$\frac{\partial f}{\partial z} = -5e^{3x+4y} \sin 5z$$

$$\frac{\partial^2 f}{\partial x^2} = 9 e^{3x+4y} \cos 5z$$

$$\frac{\partial^2 f}{\partial y^2} = 16 e^{3x+4y} \cos 5z$$

$$\frac{\partial^2 f}{\partial z^2} = -25 e^{3x+4y} \cos 5z$$

So, $\left\{ \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0 \right\}$

Note: Let $f(x, y, z)$ ($f(x, y)$) be a function of three variables (two variables) f is said to satisfy the three dimensional (two dimensional)

Laplace Equation if

$$\left\{ \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} + \frac{\partial^2 f}{\partial z^2} = 0 \right\}$$

$$\left(\frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2} = 0 \right)$$

Ex:

$$f(x, y) = \sin(xy)$$

Find $\frac{\partial^2 f}{\partial x \partial x}$ and $\frac{\partial^2 f}{\partial x \partial y}$

$$\frac{\partial f}{\partial x} = y \cos(xy)$$

$$\frac{\partial f}{\partial y} = x \cos(xy)$$

$$\frac{\partial^2 f}{\partial y \partial x} = yx \left(-\sin(xy) \right) + \cos(xy)$$

$$= \boxed{\cos(xy) - xy \sin(xy)}$$

$$\frac{\partial^2 f}{\partial x \partial y} = yx \left(-\sin(xy) \right) + \cos(xy)$$

$$= \boxed{\cos(xy) - xy \sin(xy)}$$

Ex: Show that $w_{xy} = w_{yx}$

if $w = e^x + x \ln y + y \ln x$

$$w_x = e^x + \ln y + \frac{y}{x}$$

$$w_y = \frac{x}{y} + \ln x$$

$$\boxed{w_{xy} = \frac{1}{y} + \frac{1}{x}}$$

$$\boxed{w_{yx} = \frac{1}{y} + \frac{1}{x}}$$

$$\boxed{w_{xy} = w_{yx}}$$

Ex: $f(x, y) = xy \frac{x^2 - y^2}{x^2 + y^2}$ for $(x, y) \neq (0, 0)$

$$= 0 \quad \text{for } (x, y) = (0, 0)$$

For $(x, y) \neq (0, 0)$,

$$f_x(x, y) = y(1) \frac{x^2 - y^2}{x^2 + y^2} + xy \frac{2x(x^2 + y^2) - (x^2 - y^2)2x}{(x^2 + y^2)^2}$$
$$= y \frac{x^2 - y^2}{x^2 + y^2} + \frac{4x^2 y^3}{(x^2 + y^2)^2}$$

$$\text{and } f_y(x, y) = x(1) \frac{x^2 - y^2}{x^2 + y^2} + xy \frac{(-2y)(x^2 + y^2) - (x^2 - y^2)2y}{(x^2 + y^2)^2}$$
$$= x \frac{x^2 - y^2}{x^2 + y^2} - \frac{4x^3 y^2}{(x^2 + y^2)^2}$$

$$\text{Hence } f_x(0, h) = h \frac{0^2 - h^2}{0^2 + h^2} + 0 = h \left(\frac{-h^2}{h^2} \right) = \boxed{-h}$$

$$\text{and } f_y(h, 0) = h \frac{h^2 - 0^2}{h^2 + 0^2} - 0 = h \left(\frac{h^2}{h^2} \right) = \boxed{h}$$

$$\text{Now } f_x(0, 0) = \lim_{h \rightarrow 0} \frac{f(0+h, 0) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f(h, 0) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{0 \times \frac{h^2}{h^2} - 0}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = \boxed{0}$$

$$f_y(0, 0) = \lim_{h \rightarrow 0} \frac{f(0, 0+h) - f(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{f(0, h) - f(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{0 \times \left(\frac{-h^2}{h^2} \right) - 0}{h} = \lim_{h \rightarrow 0} \frac{0}{h} = \boxed{0}$$

$$\text{Now } f_{xy}(0, 0) = \lim_{h \rightarrow 0} \frac{f_x(0, 0+h) - f_x(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f_x(0, h) - f_x(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{-h - 0}{h} = \lim_{h \rightarrow 0} \frac{-h}{h} = \lim_{h \rightarrow 0} (-1) = -1$$

So, $\boxed{f_{xy}(0, 0) = -1}$

$$f_{yx}(0, 0) = \lim_{h \rightarrow 0} \frac{f_y(0+h, 0) - f_y(0, 0)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{f_y(h, 0) - f_y(0, 0)}{h} = \lim_{h \rightarrow 0} \frac{h - 0}{h}$$

$$= \lim_{h \rightarrow 0} \frac{h}{h} = \lim_{h \rightarrow 0} 1 = 1$$

So, $\boxed{f_{yx}(0, 0) = 1}$

Therefore in this case

$$\boxed{f_{xy}(0, 0) \neq f_{yx}(0, 0)}$$

Euler's Theorem on Mixed Derivatives

If $f(x, y)$ and its partial derivatives f_x , f_y , f_{xy} and f_{yx} are defined in an open region containing a point (a, b) and are all continuous at (a, b) , then $\boxed{f_{xy}(a, b) = f_{yx}(a, b)}$

Homogeneous function:

$f(x, y)$ is called a homogeneous function of order n if

$$f(tx, ty) = t^n f(x, y).$$

or equivalently

$$f(x, y) = x^n F(y/x)$$

• For $f(x, y, z)$: $f(tx, ty, tz) = t^n f(x, y, z)$

• For $f(x_1, x_2, \dots, x_m)$: $f(tx_1, tx_2, \dots, tx_m) = t^n f(x_1, x_2, \dots, x_m)$

Ex: $f(x, y, z) = Ax^3 + By^3 + Cz^3 + Dxy^2 + Exz^2 + Gyx^2 + Hxz^2 + Izy^2 + Jxyz$

is a homogeneous function of order 3

Ex: Prove that $f(x, y) = x^3 - 2x^2y + 3xy^2 + y^3$

is a homogeneous function of order 3

$$f(tx, ty) = (tx)^3 - 2(tx)^2(ty) + 3(tx)(ty)^2 + (ty)^3 \\ = t^3 [x^3 - 2x^2y + 3xy^2 + y^3] = t^3 f(x, y)$$

So, f is a homogeneous function of order 3.

$$\text{Now } \left. \begin{array}{l} \frac{\partial f}{\partial x} = 3x^2 - 4xy + 3y^2 \\ \frac{\partial f}{\partial y} = -2x^2 + 6xy + 3y^2 \end{array} \right\}$$

$$\text{Then } x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 3x^3 - 4x^2y + 3xy^2 - 2x^2y + 6xy^2 + 3y^3 \\ = 3x^3 - 6x^2y + 9xy^2 + 3y^3 \\ = 3[x^3 - 2x^2y + 3xy^2 + y^3] = 3f(x, y)$$

$$\text{Thus } \boxed{x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} = 3f(x, y)}$$

Euler's Theorem for Homogeneous function

If $f(x, y, z)$ is a homogeneous function of order n then

$$\boxed{x \frac{\partial f}{\partial x} + y \frac{\partial f}{\partial y} + z \frac{\partial f}{\partial z} = nf}$$

So, for n variables,

$$x_1 \frac{\partial f}{\partial x_1} + x_2 \frac{\partial f}{\partial x_2} + \dots + x_m \frac{\partial f}{\partial x_m} = nf$$

Ex ① If $u(x, y) = \frac{x^2 + y^2}{\sqrt{x+y}}$

Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = \frac{3}{2}u$

Ex ②: If $u(x, y) = \ln \left(\frac{x^2 + y^2}{x+y} \right)$

Prove that $x \frac{\partial u}{\partial x} + y \frac{\partial u}{\partial y} = 1$

Ex ③: If $W = \ln \left(\frac{5x^3y^4 + 7y^2xz^4 - 75y^3z^4}{x^2 + y^2} \right)$

find $x \frac{\partial W}{\partial x} + y \frac{\partial W}{\partial y} + z \frac{\partial W}{\partial z}$ (Answer: 5)

Chain Rule :

Recall for one variable

If f and g are real valued functions in one variable and if $g \circ f$ is defined in some neighborhood composition of two functions

of x_0 .

Now if f is differentiable at x_0

and g is differentiable at $f(x_0)$

then $g \circ f$ is differentiable at x_0

and
$$(g \circ f)'(x_0) = g'(f(x_0)) f'(x_0)$$

Equivalently

If $W = f(x)$ (differentiable)

and $x = x(t)$ (differentiable),

then W is a differentiable function of t and

$$\frac{dW}{dt} = \frac{dW}{dx} \cdot \frac{dx}{dt}$$

Generalization:

(I) If $W = f(x, y)$ is differentiable, and x and y are differentiable function of t , then W is a

differentiable function of t

$$\boxed{\frac{dW}{dt} = \frac{\partial f}{\partial x} \cdot \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt}}$$

(II) If $W = f(x, y, z)$ is differentiable and x, y, z are differentiable function of t , then W is a differentiable function of t

$$\boxed{\frac{dW}{dt} = \frac{\partial f}{\partial x} \frac{dx}{dt} + \frac{\partial f}{\partial y} \frac{dy}{dt} + \frac{\partial f}{\partial z} \frac{dz}{dt}}$$

Ex: Let $W = \ln(x^2 + y^2 + z^2)$

$$x = \cos t, \quad y = \sin t, \quad z = 4\sqrt{t}$$

Find $\frac{dW}{dt}$ at $t = 3$.

$$\begin{aligned} \frac{dW}{dt} &= \frac{2x}{x^2 + y^2 + z^2} (-\sin t) + \frac{2y}{x^2 + y^2 + z^2} (\cos t) \\ &\quad + \frac{2z}{x^2 + y^2 + z^2} 4 \times \frac{1}{2} t^{-1/2} \end{aligned}$$

$$\stackrel{= \dots =}{\text{(simplifying)}} \boxed{\frac{16}{1+16t}}$$

Now, at $t=3$,

$$\left. \frac{dW}{dt} \right|_{t=3} = \boxed{\frac{16}{49}}$$

Σx_i Let $W = f(x_1, \dots, x_n)$

and $x_i = x_i(t)$ for $i=1, 2, \dots, n$

Assume differentiability for all the functions.

Then $\frac{dW}{dt} = ?$

Step III: Let $W = f(x, y)$

Let $\left. \begin{aligned} x &= g(r, s) \\ y &= h(r, s) \end{aligned} \right\}$

Assume that all the functions are differentiable.

Then

$$\boxed{\frac{\partial W}{\partial r} = \frac{\partial W}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial W}{\partial y} \frac{\partial y}{\partial r}}$$

and

$$\boxed{\frac{\partial W}{\partial s} = \frac{\partial W}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial W}{\partial y} \frac{\partial y}{\partial s}}$$

Step IV: Let $W = f(x, y, z)$ where $x = g(r, s)$, $y = h(r, s)$
and $z = k(r, s)$

If all the functions are differentiable,

then

$$\boxed{\frac{\partial W}{\partial r} = \frac{\partial W}{\partial x} \frac{\partial x}{\partial r} + \frac{\partial W}{\partial y} \frac{\partial y}{\partial r} + \frac{\partial W}{\partial z} \frac{\partial z}{\partial r}}$$

$$\boxed{\frac{\partial W}{\partial s} = \frac{\partial W}{\partial x} \frac{\partial x}{\partial s} + \frac{\partial W}{\partial y} \frac{\partial y}{\partial s} + \frac{\partial W}{\partial z} \frac{\partial z}{\partial s}}$$

Ex: ① $z = \tan^{-1}\left(\frac{x}{y}\right)$ } Find $\frac{\partial z}{\partial r}$ and $\frac{\partial z}{\partial \theta}$
 $x = r \cos \theta$
 $y = r \sin \theta$ } at $(1.3, \frac{\pi}{6})$

Ex ②: $u = e^{qr} \sin^{-1}(p)$

where $p = \sin x$, $q = z^2 \ln y$

and $r = \frac{1}{z}$

Find

$$\frac{\partial u}{\partial x}, \quad \frac{\partial u}{\partial y} \quad \text{and} \quad \frac{\partial u}{\partial z}$$

$$\text{at} \left(\frac{\pi}{4}, \frac{1}{2}, -\frac{1}{2} \right)$$

$$\frac{\partial u}{\partial x} = \frac{\partial u}{\partial p} \frac{\partial p}{\partial x} + \frac{\partial u}{\partial q} \frac{\partial q}{\partial x} + \frac{\partial u}{\partial r} \frac{\partial r}{\partial x}$$

$$= \frac{e^{qr}}{\sqrt{1-p^2}} \cos x + 0 + 0$$

$$= \frac{e^{qr} \cos x}{\sqrt{1-p^2}} = \frac{e^{\frac{z^2 \ln y}{2}} \cos x}{\sqrt{1-\sin^2 x}}$$

$$= e^{\frac{z^2 \ln y}{2}} \frac{\cos x}{\cos x} = e^{\frac{z^2}{2}} = \boxed{y^{\frac{z^2}{2}}} \quad \left(\text{since } -\frac{\pi}{2} < x < \frac{\pi}{2} \right)$$

$$\text{At} \left(\frac{\pi}{4}, \frac{1}{2}, -\frac{1}{2} \right)$$

$$\frac{\partial u}{\partial x} = \left(\frac{1}{2} \right)^{-1/2} = \boxed{\sqrt{2}}$$

Similarly find $\frac{\partial u}{\partial y}$ and $\frac{\partial u}{\partial z}$ (H.W.)